

FIITJEE NSEC MOCK TEST-5

Time Allotted: 2 Hours

CODE:

Maximum Marks: 216

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

Write the question paper code (mentioned above) on YOUR OMR Answer Sheet (in the space provided), otherwise your Answer Sheet will NOT be evaluated. Note that the same Question Paper Code appears on each page of the question paper.

Instructions to Candidates:

1. Use of mobile phone, smart watch, and ipad during examination is STRICTLY PROHIBITED.
2. In addition to this question paper, you are given OMR Answer Sheet along with Candidate's copy.
3. On the OMR sheet, make all the entries carefully in the space provided **ONLY** in **BLOCK CAPITALS** as well as by properly darkening the appropriate bubbles.

Incomplete/ incorrect/ carelessly filled information may disqualify your candidature.

4. On the OMR Answer Sheet, use only **BLUE** or **BLACK BALLPOINT PEN** for making entries and filling the bubbles.
5. **Your Enrollment number entered** on the OMR Answer Sheet shall remain your login credentials means login id and password respectively for accessing your performance / result in NSE.
6. Question paper has two parts. In part A1 (Q. No.1 to 48) each question has four alternatives, out of which only one is correct. Choose the correct alternative (s) and fill the appropriate bubble (s), as shown.

Q. No. 22	☐ a	☒	☐ c	☐ d
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In part A2 (Q. No. 49 to 60) each question has four alternatives out of which any number of alternative (s) (1, 2, 3, or 4) may be correct. You have to choose all correct alternative(s) and fill the appropriate bubble(s), as shown

Q. No. 54	☐ a	☒	☐ c	☒
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7. For **Part A1**, each correct answer carries 3 marks whereas 1 mark will be deducted for each wrong answer. In **Part A2**, you get 6 marks if all the correct alternatives are marked. No negative marks in this part.
8. Rough work may be done in the space provided. There are 15 printed pages in this paper
9. Use of **non - programmable scientific** calculator is allowed.
10. No candidate should leave the examination hall before the completion of the examination.
11. After submitting answer paper, take away the question paper & Candidate's copy OMR sheet for your reference.

Please DO NOT make any mark other than filling the appropriate bubbles properly in the space provided on the OMR answer sheet.

OMR answer sheets are evaluated using machine, hence CHANGE OF ENTRY IS NOT ALLOWED. Scratching or overwriting may result in a wrong score.

DO NOT WRITE ON THE BACK SIDE OF THE OMR ANSWER SHEET.

Name of the Candidate : _____

Batch : _____ Date of Examination : _____

Enrolment Number : _____

**INDIAN ASSOCIATION OF PHYSICS TEACHERS
 NATIONAL STANDARD EXAMINATION IN CHEMISTRY
 (NSEC-2023)**

Time: 120 minutes

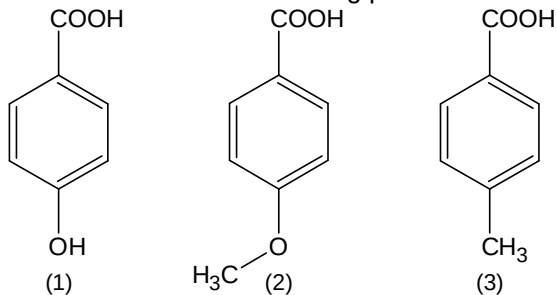
Max. Marks: 216

Attempt All Sixty Questions

A – 1

ONLY ONE OUT OF FOUR OPTIONS IS CORRECT BUBBLE THE CORRECTION OPTION

1. The correct order of increasing pKa values is



- (A) $3 < 2 < 1$
 (C) $3 < 1 < 2$

- (B) $1 < 2 < 3$
 (D) $1 < 3 < 2$

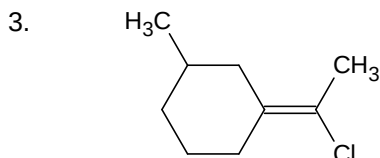
2. List – 1 (Reaction)

- (a) Riemer Tiemann reaction
 (b) Pinacol Pinacolone rearrangement
 (c) Hofmann bromamide reaction
 (d) Perkins reaction

- | | a | b | c | d |
|-----|---|---|---|---|
| (A) | 2 | 1 | 3 | 4 |
| (B) | 2 | 3 | 1 | 4 |
| (C) | 3 | 1 | 2 | 4 |
| (D) | 1 | 2 | 3 | 4 |

List – 2 (intermediates)

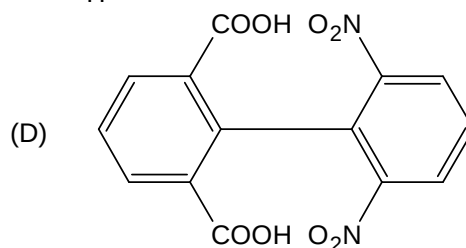
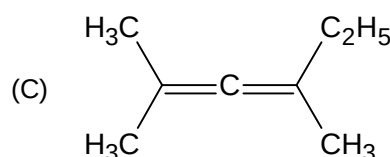
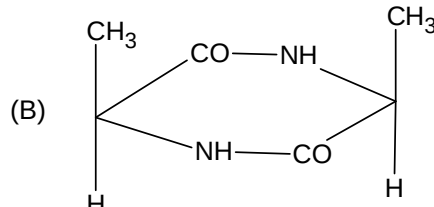
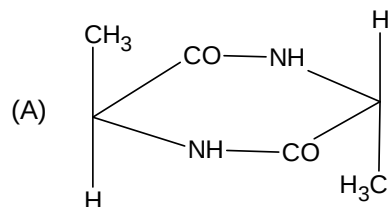
- (1) Isocyanate
 (2) dichlorocarbene
 (3) Carbocation
 (4) Carbanion



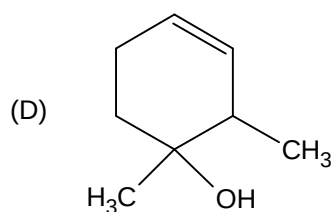
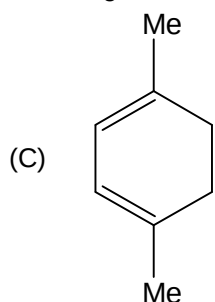
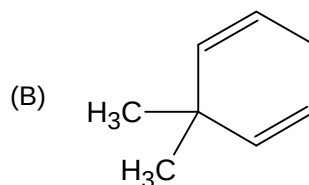
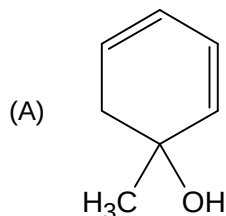
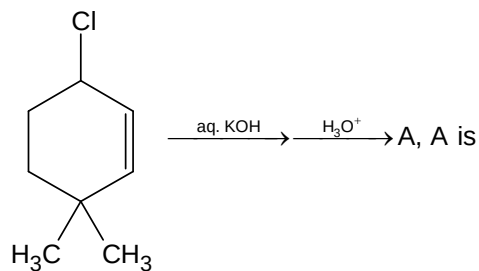
Which of the following statements is true about this compound?

- (A) It exhibit only optical isomerism
 (B) it exhibit only geometrical isomerism
 (C) It exhibit both geometrical and optical isomerism
 (D) none of the above in

4. Which of the following compound is resolvable?

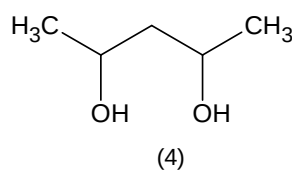
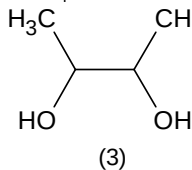
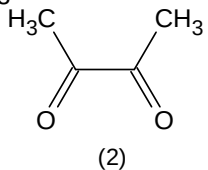
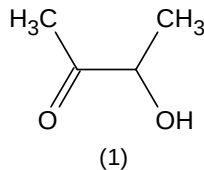


5.



6.

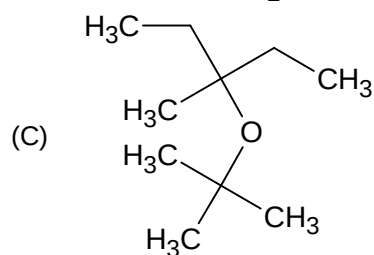
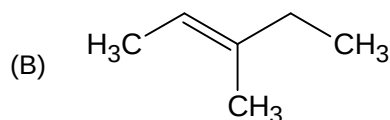
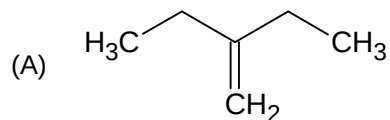
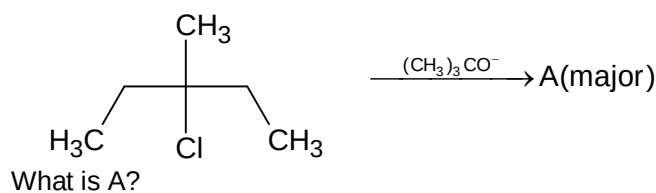
Which of the following will be oxidized by HIO_4 ?



(A) 1, 2, 3 and 4
(C) 1, 2, and 3

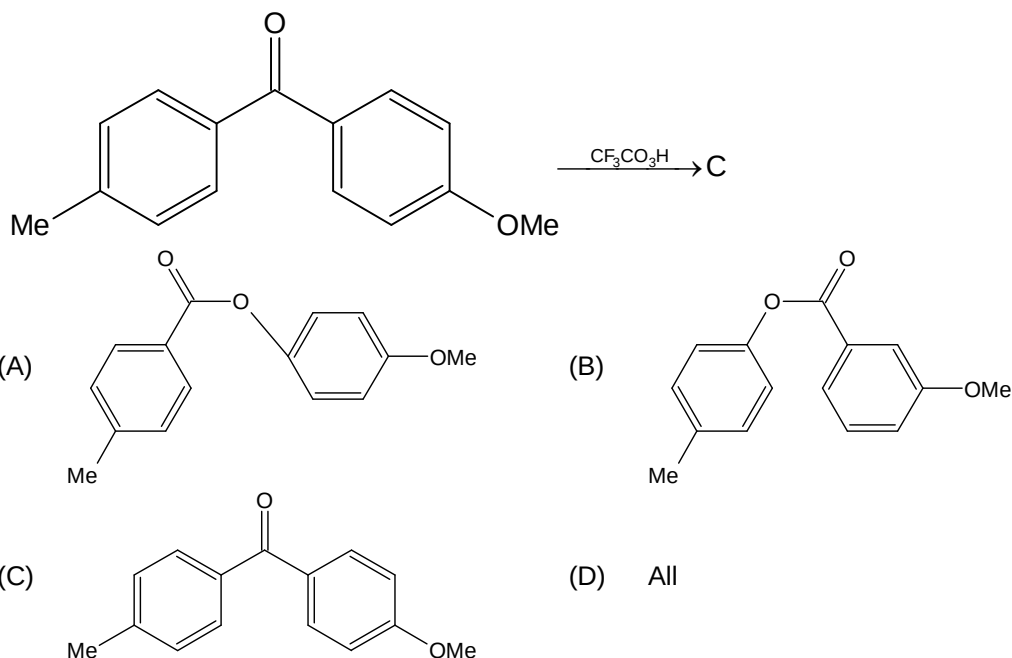
(B) 1 & 3
(D) 2 and 3

7.

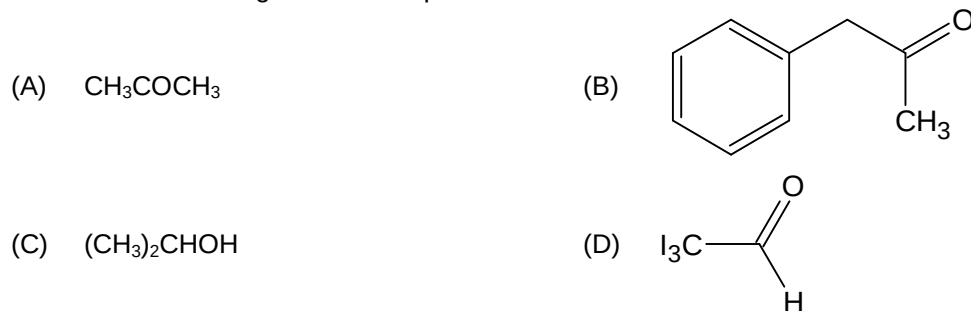


(D) None

8.



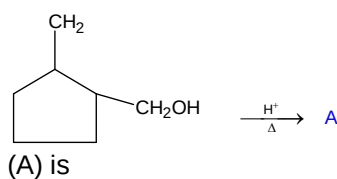
9. Which of the following does not respond to the iodoform test?



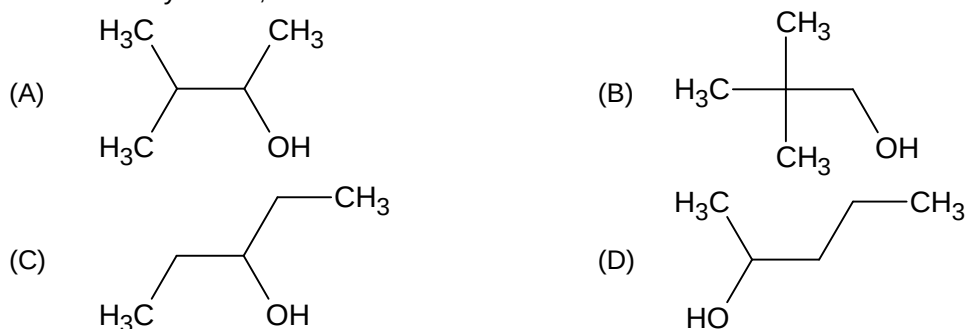
10. Oxidation of CH_3CHO by SeO_2 proceeds through

- (A) aldehyde form of CH_3CHO
 (B) hydrated form of CH_3CHO
 (C) enol form of CH_3CHO
 (D) none of these

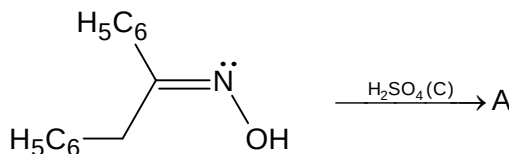
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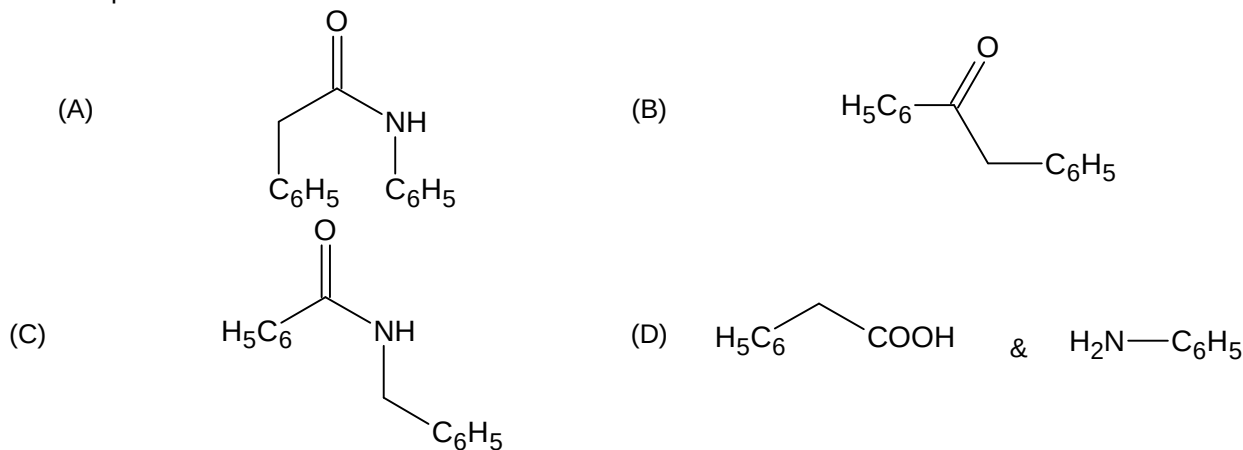
12. An alcohol $C_5H_{11}OH$ on dehydration gives an alkene which on oxidation yields a mixture of a ketone and a carboxylic acid, the alcohol is



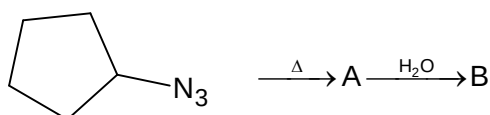
13.



Compound A is



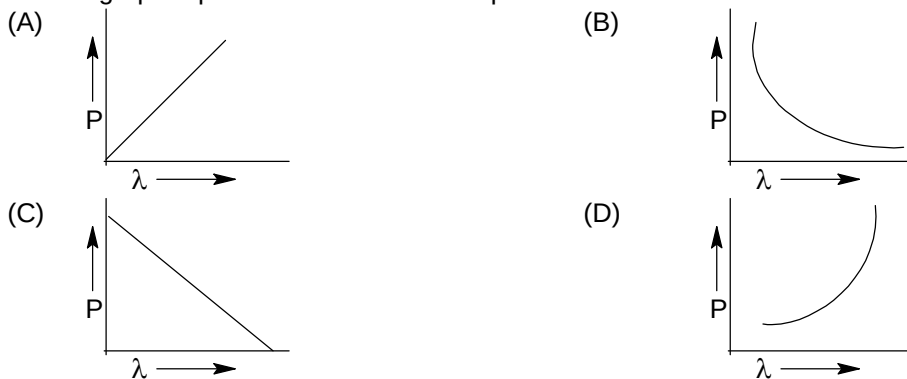
14.



The product B

- (A) Gives positive Tollen's test but does not react with $CHCl_3/KOH$
 (B) Gives positive isocyanide test but does not perform Tollen's Test
 (C) Does not perform both tests
 (D) Gives positive Tollen's test as well as isocyanide test

15. Which graph represents the variation of photon momentum 'P' with wavelength λ ?

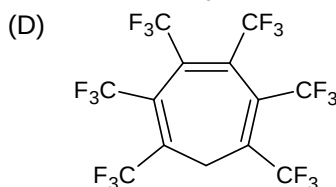
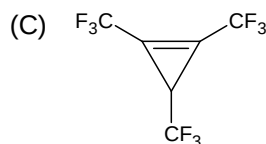
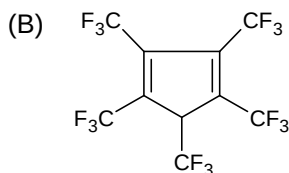
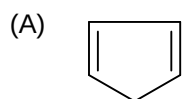


16. What is the uncertainty in position of an electron (mass = 9.1×10^{-31} Kg) moving at 40% of the velocity of light with a velocity uncertainty of 0.10%?
 (A) 6×10^{-4} (B) 6×10^{-9}
 (C) 36×10^{-8} (D) 36×10^{-18}

17. $2[\text{Fe}(\text{CN})_6]^{3-} + 2\text{OH}^- + \text{H}_2\text{O}_2 \rightarrow 2[\text{Fe}(\text{CN})_6]^{4-} + 2\text{H}_2\text{O} + \text{O}_2$ is
 (A) H_2O_2 is an oxidant
 (B) CN^- undergoes neither oxidation nor reduction.
 (C) $2[\text{Fe}(\text{CN})_6]^{3-}$ liberates electrons
 (D) all the above

18. The compound behaving only as oxidising agent is
 (A) H_2O_2 (B) SO_2
 (C) HNO_2 (D) HNO_3

19. Which of the following species is highly acidic?



20. List – 1 (ionic species)

- (a) XeF_5^+
 (b) SiF_5^-
 (c) AsF_4^+
 (d) ICl_4^-

- List – 2 (shapes)

- (1) Tetrahedral
 (2) Square planer
 (3) Trigonal bipyramidal
 (4) Square pyramidal
 (5) Octahedral

	a	b	c	d
(A)	1	2	5	3
(B)	1	3	5	2
(C)	4	2	1	3
(D)	4	3	1	2

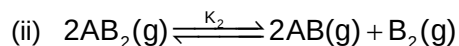
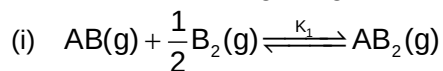
21. The increasing order of thermal stability for following salts will be

- (A) $\text{FeC}_2\text{O}_4 < \text{CaC}_2\text{O}_4 < \text{CaCO}_3 < \text{K}_2\text{C}_2\text{O}_4$
 (B) $\text{K}_2\text{C}_2\text{O}_4 < \text{CaCO}_3 < \text{FeC}_2\text{O}_4 < \text{CaC}_2\text{O}_4$
 (C) $\text{FeC}_2\text{O}_4 < \text{K}_2\text{C}_2\text{O}_4 < \text{CaC}_2\text{O}_4 < \text{CaCO}_3$
 (D) $\text{CaCO}_3 < \text{K}_2\text{C}_2\text{O}_4 < \text{CaC}_2\text{O}_4 < \text{FeC}_2\text{O}_4$

22. Which of the following compound have octahedral structure and zero lone pair?

- (A) XeO_4 (B) $\text{Ba}_2[\text{XeO}_6]$
 (C) XeOF_4 (D) XeO_2F_2

23. Consider the following two gaseous equilibria at a definite temperature.



If K_1 & K_2 are the equilibrium constants for the reactions (i) & (ii) respectively then the relation between K_1 & K_2 is

(A) $K_1 \cdot K_2 = 1$

(B) $K_1^2 = K_2$

(C) $K_1^2 \cdot K_2 = 1$

(D) $K_1^2 = K_2(0.5)$

24. A first order reaction has a half – life of 14.5 hours. What percentage of the reactant will remain after 24.0 hours ?

(A) 18.3%

(B) 31.8%

(C) 45.5%

(D) 68.2%

25. Bicyclohexane was found to undergo two parallel 1st order rearrangements. At 730 K the first order rate constant for the formation of cyclohexene was $1.26 \times 10^{-4} \text{ s}^{-1}$ and for the formation of methyl cyclopentene the rate constant was $3.8 \times 10^{-5} \text{ s}^{-1}$. What was the percentage distribution of methyl cyclopentene.

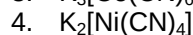
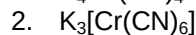
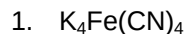
(A) 77%

(B) 23%

(C) 66%

(D) 34%

26. Which of the following are diamagnetic?



Select the correct choice using the codes given below

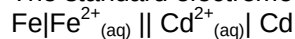
(A) 1, 2 and 4

(B) 1, 3 and 4

(C) 2 and 3

(D) 1 and 4

27. The standard electromotive force of the cell



If the temperature co-efficient of emf is -0.125 V K^{-1} , the value of ΔS° at 25°C would be

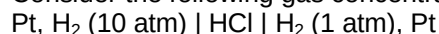
(A) $-20.125 \text{ KJ K}^{-1}$

(B) $-24.125 \text{ KJ K}^{-1}$

(C) $-26.125 \text{ KJ K}^{-1}$

(D) $-22.125 \text{ KJ K}^{-1}$

28. Consider the following gas concentration cell



The emf of the above cell is

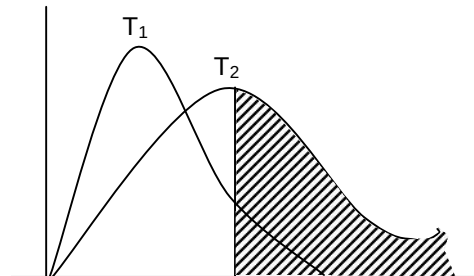
(A) 0.0295 V

(B) 0.0591 V

(C) 0.1182 V

(D) 0.2364 V

29. The distribution of molecular kinetic energy at two temp. is as shown in the following graph



Which of the following condition is correct?

(A) The number of molecules with energy E_a or greater is proportional to the shaded area for each temperature.

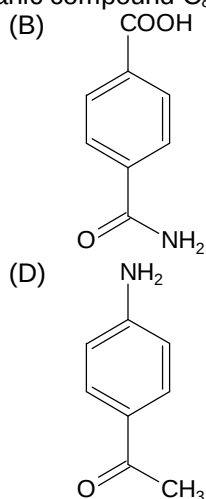
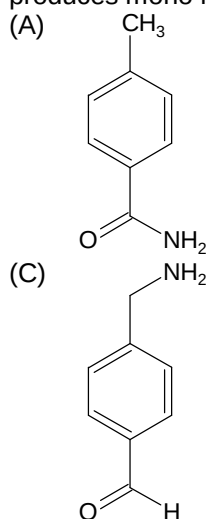
(B) The number of molecules with energy E_a or less is proportional to the shaded area for each temperature.

(C) The number of molecules with energy E_a is the sum of all temperature.

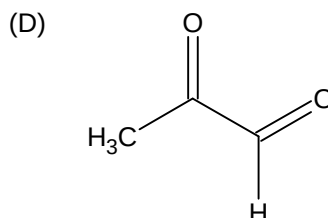
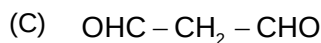
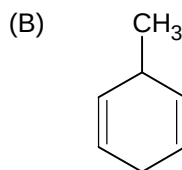
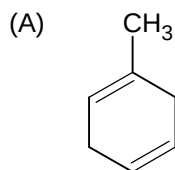
(D) The graph does not follows Maxwell Boltzmann energy distribution law.

30. A 10 L cylinder of oxygen at 4 atm pressure and 17°C developed a leak. When the leak was repaired, 2.5 atm of O₂ remained in the cylinder, still at 17°C. The no. of moles of gas escaped.
 (A) 1.2 mol (B) 0.30 mol
 (C) 0.63 mol (D) 2.4 mol
31. A 0.10 M solution of F⁻ is gradually added to a solution containing Ba²⁺, Ca²⁺ & Pb²⁺ ions, each at a concentration of 1×10⁻³ M. In what order from first to last, will the precipitates of BaF₂, CaF₂, PbF₂ form.
 $K_{sp}(\text{BaF}_2) = 1.8 \times 10^{-7}$; $K_{sp}(\text{CaF}_2) = 1.5 \times 10^{-10}$
 $K_{sp}(\text{PbF}_2) = 7.1 \times 10^{-7}$
 (A) CaF₂, PbF₂, BaF₂ (B) BaF₂, CaF₂, PbF₂
 (C) PbF₂, BaF₂, CaF₂ (D) CaF₂, BaF₂, PbF₂
32. The volume to which 50 ml 6.0 M ammonia solution be diluted to double the degree of protonation is (K_b = 2×10⁻⁵)
 (A) 220 ml (B) 200 ml
 (C) 400 ml (D) 240 ml
33. Insulin (C₂H₁₀O₅)_n is dissolved in a suitable solvent and the osmotic pressure (π) of solution of various concentration Kg/m³ is measured at 20°C. The slope of a plot of π against C is found to be 4.65×10⁻³ (SI units). The molecular weight of the insulin (in Kg) is
 (A) 4.8×10⁵ (B) 9×10⁵
 (C) 3×10⁵ (D) 5.15×10⁵
34. A₁ g of an element give A₂ g of its oxide. The equivalent mass of the element is
 (A) $\frac{A_2 - A_1}{A_1} \times 8$ (B) $\frac{A_2 - A_1}{A_2} \times 8$
 (C) $\frac{A_1}{A_2 - A_1} \times 8$ (D) (A₂ - A₁)×8
35. A freshly prepared radioactive source of half life 2 hours emits radiation of intensity which is 64 times the permissible safe value. The minimum time after which it would be possible to work safely with the source is:
 (A) 6hr (B) 12 hr
 (C) 24 hr (D) 128 hr
36. Sodium thiosulphate is formed when
 (A) A solution of Na₂SO₃ is heated with finely powdered sulphur
 (B) NaHSO₃ is treated with NaHS
 (C) I₂ is added to a mixture of Na₂S and Na₂SO₃
 (D) All the above
37. Which one of the following oxoanion contain S – S bond?
 (1) S₂O₈⁻², (2) S₂O₃⁻², (3) S₂O₄⁻², (4) S₂O₆⁻²
 Select the correct answers using codes below
 (A) 1, 2 & 3 (B) 1,3 & 4
 (C) 2, 3 & 4 (D) 2 & 4
38. Borax bead test is not given by
 (A) an aluminium salt (B) a cobalt salt
 (C) a copper salt (D) a nickel salt
39. A compound which can be used in space vehicles both to absorb CO₂ and liberate O₂ is
 (A) NaOH (B) NaO
 (C) Na₂O₂ (D) CaO & NaOH

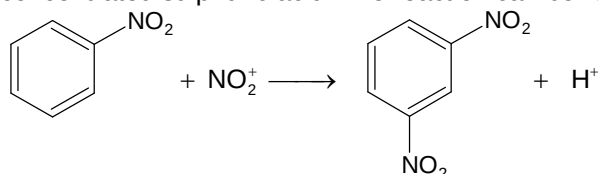
40. Mg_2C_3 has the following characteristics
 (A) It is called magnesium allyide (B) It contains Mg^{2+} and C_3^{-4} ions
 (C) It on hydrolysis gives propyne (D) All of the above
41. A compound alloy of gold and copper crystallizes in a cubic lattice in which the gold atoms occupy the lattice points at the corners of a cube and the copper atoms occupy the centres of each of the cube faces. The formula of this compound is
 (A) AuCu (B) Au_2Cu
 (C) Au_3Cu (D) AuCu_3
42. 0.7 g of $\text{Na}_2\text{CO}_3 \cdot x\text{H}_2\text{O}$ were dissolved in water and the volume was made to 100 mL, 20 mL of this solution required 19.8 mL of N/10 HCl for complete neutralization. The value of x is
 (A) 7 (B) 2
 (C) 3 (D) 5
43. A 3 mol sample of a triatomic ideal gas at 300 K is allowed to expand under isobaric adiabatic condition from 5 L to 40 L. The value of ΔH is
 (A) -12.46 KJ (B) -14.965 KJ
 (C) -24.62 KJ (D) -10.24 KJ
44. Find the work done when 18 gm water vaporises at 100°C . Density of water at this temperature is 1 gm/ml
 (A) 1.006 KJ (B) 4.63 KJ
 (C) 3.058 KJ (D) 4.938 KJ
45. The major role of fluorspar (CaF_2) which is added in small quantities in the electrolytic reduction of alumina dissolved in fused cryolite (Na_3AlF_6) is
 (A) as a catalyst
 (B) to make the fused mix very conducting
 (C) to lower the temperature of melt and increase the conductivity of fused mix
 (D) to decrease the rate of oxidation of carbon at the anode
46. An organic compound $\text{C}_8\text{H}_9\text{NO}$ is found to be insoluble in dilute acid and base. On treatment with $\text{KMnO}_4/\text{H}_2\text{SO}_4$, the compound formed is devoid of nitrogen and is soluble in NaHCO_3 and on nitration produces mono nitrosubstituted product, the organic compound $\text{C}_8\text{H}_9\text{NO}$ is



47. In the following reaction,



48. 1, 3-dinitrobenzene can be prepared by heating nitrobenzene with a mixture of fuming nitric acid and concentrated sulphuric acid. The reaction can be represented by the following equation.



If the yield of the reaction is 55%, the mass of 1, 3-dinitrobenzene produced from 12.30 g of nitrobenzene is

(A) 16.90 g

(B) 16.80 g

(C) 9.30 g

(D) 9.24 g

ANY NUMBER OF OPTIONS 4, 3, 2 or 1 MAY BE CORRECT

MARKS WILL BE AWARDED ONLY IF ALL THE CORRECT OPTIONS ARE BUBBLED

49. Mixture of $\text{C}_6\text{H}_5\text{CHO}$ and HCHO is treated with NaOH then Cannizzaro reaction involves

(A) oxidation of HCHO

(B) reduction of HCHO

(C) oxidation of $\text{C}_6\text{H}_5\text{CHO}$

(D) reduction of $\text{C}_6\text{H}_5\text{CHO}$

50. Nucleophilic substitution in alkyl halides

(A) is an important reaction of organo-halogen compounds

(B) is possible because of the electronegativity difference of carbon atom

(C) takes place since carbon atom has tendency to attract negatively charged or electron rich groups such as OH^- , NH_3 etc

(D) takes place because of the polar nature of carbon-halogen bond

51. An optically active alcohol (A) $[\text{C}_6\text{H}_{10}\text{O}]$ absorbs two moles of hydrogen per mole of [A] upon catalytic hydrogenation and gives a product (B). (B) is resistant to oxidation by CrO_3 and does not show any optical activity. What is not true regarding (A) and (B).

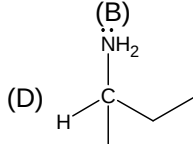
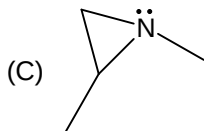
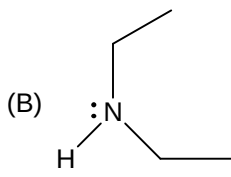
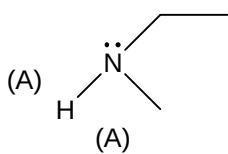
(A) (B) is optically active

(B) (A) can be easily oxidized

(C) (B) shows +ve iodoform test

(D) (A) shows -ve iodoform test

52. Which of the following amines can be resolved into two enantiomers?



53. For radial probability distribution curves, which of the following is/are correct?

- (A) The number of maxima in 2s orbital are two
 (B) The number of spherical or radial nodes is equal to $n - \ell - 1$
 (C) The number of angular nodes are ' ℓ '
 (D) $3d_z^2$ has two angular nodes

54. Which of the following have undistorted octahedral structures?

- (A) SF_6 (B) PF_6^-
 (C) SiF_6^{2-} (D) XeF_6

55. The equation $\alpha = \frac{D - d}{(n - 1)d}$ is not correctly matched for

where D = vapour density of 'A' before dissociation, d = V.D. of mixture.

- (A) $A \rightleftharpoons \left(\frac{n}{2}\right)B + \left(\frac{n}{3}\right)C$ (B) $A \rightleftharpoons \left(\frac{n}{3}\right)B + \left(\frac{2n}{3}\right)C$
 (C) $A \rightleftharpoons \left(\frac{n}{2}\right)B + \left(\frac{n}{4}\right)C$ (D) $A \rightleftharpoons \left(\frac{n}{2}\right)B + C$

56. Arrhenius equation is represented as: $k = Ae^{-E_a/RT}$. Which of the following options is/are correct?

- (A) Graph plotted between k and 1/T is a straight line
 (B) Graph plotted between log k and 1/T is a straight line with a positive slope
 (C) Intercept of the graph between log k and 1/T = log A
 (D) Slope of the graph between log k and 1/T = $-2.303E_a/R$

57. An acid-base indicator has a K_a of 3.0×10^{-5} . The acid form of the indicator is red and the basic form is blue. Then

- (A) pH is 4.05 when indicator is 75% red (B) pH is 5.00 when indicator is 75% blue
 (C) pH is 5.00 when indicator is 75% red (D) pH is 4.05 when indicator is 75% blue

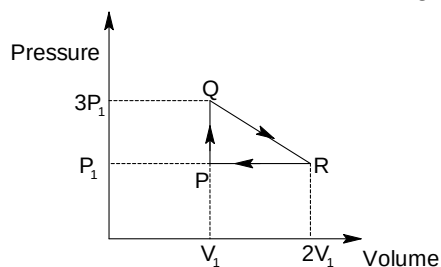
58. Which of the following substance is/are soluble in hot caustic soda containing an excess of concentrated solution of Br_2 water?

- (A) $\text{Cr}(\text{OH})_3$ (B) $\text{Mn}(\text{OH})_3$
 (C) $\text{Al}(\text{OH})_3$ (D) $\text{Fe}(\text{OH})_3$

59. Which of the following are outer orbital complexes?

- (A) $[\text{Co}(\text{NH}_3)_6]^{3+}$ (B) $[\text{Mn}(\text{CN})_6]^{3-}$
 (C) $[\text{Ni}(\text{NH}_3)_6]^{2+}$ (D) $[\text{FeF}_6]^{3-}$

60. One mole of an ideal monoatomic gas is taken round the cyclic process PQRP as shown in figure:



The correct statement(s) about the given cyclic process is/are (taking final magnitude only).

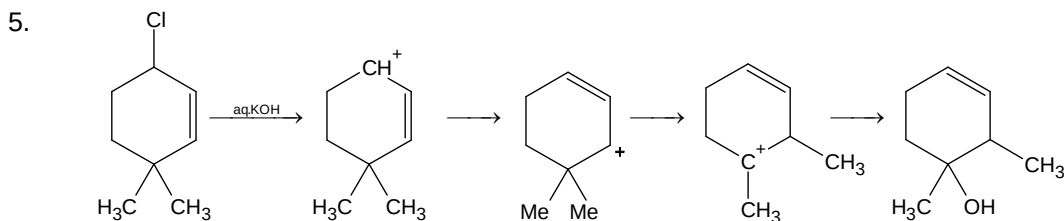
- (A) Work done by the gas is $\left(\frac{P_1 V_1}{2}\right)$
- (B) The maximum temperature attained by the gas during cycle $\left(\frac{25P_1 V_1}{8R}\right)$
- (C) The heat absorbed by path PQ is $(3P_1 V_1)$
- (D) The net heat rejected by gas in path RP is $\frac{5}{2}P_1 V_1$

ANSWERS

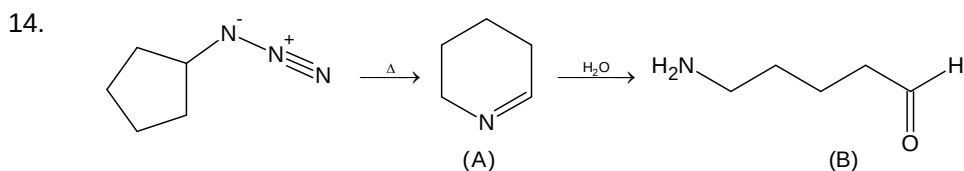
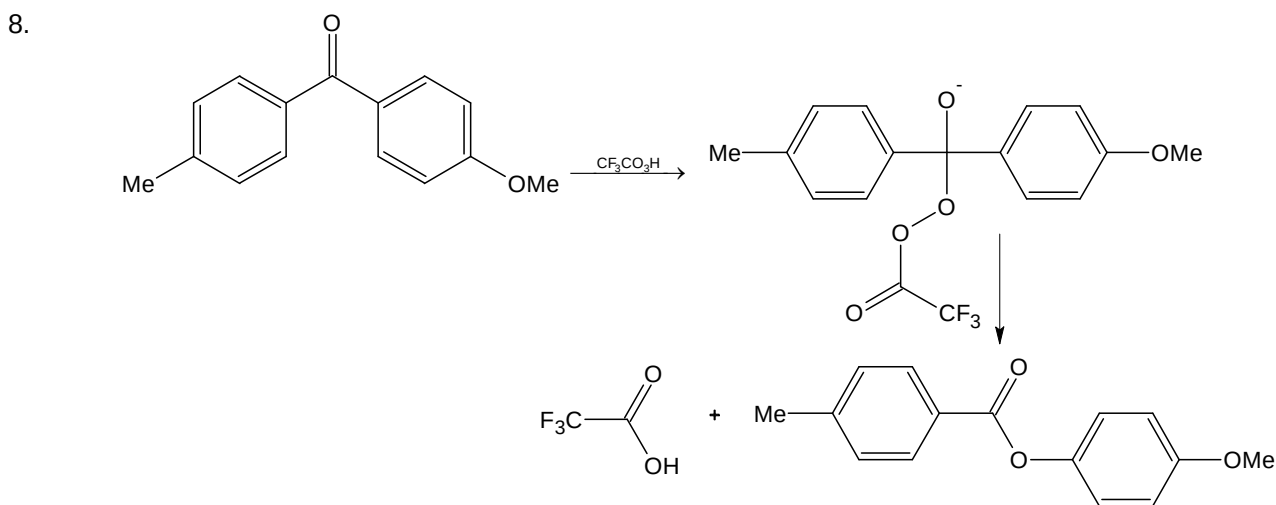
- | | | | |
|-------------|----------------|-------------|-------------|
| 1. A | 2. B | 3. C | 4. B |
| 5. D | 6. C | 7. A | 8. A |
| 9. B | 10. C | 11. C | 12. A |
| 13. A | 14. D | 15. B | 16. B |
| 17. B | 18. D | 19. B | 20. D |
| 21. A | 22. B | 23. C | 24. B |
| 25. B | 26. A | 27. B | 28. A |
| 29. A | 30. C | 31. A | 32. B |
| 33. D | 34. C | 35. B | 36. D |
| 37. C | 38. A | 39. C | 40. D |
| 41. D | 42. B | 43. B | 44. C |
| 45. C | 46. A | 47. C | 48. D |
| 49. A, D | 50. A, B, C, D | 51. A, B, C | 52. C, D |
| 53. A, B, C | 54. A, B, C | 55. A, C, D | 56. C, D |
| 57. A, B | 58. A, C | 59. C, D | 60. B, C, D |

HINTS & SOLUTIONS

4. Compound (B) don't have any symmetry



7. Base is bulky so it prefers terminal β – carbon.



As B compound contains both an aldehyde group as well as P amine have it give both tests. 15.

$$\lambda = h/p \quad \left(\lambda \propto \frac{1}{p} \right)$$

16. We, know

$$\Delta x \times m \times \Delta v \geq \frac{h}{4\pi}$$

uncertainty in velocity

$$= 0.4 \times 3 \times 10^8 \times 0.001$$

$$\Delta x \geq \frac{h}{4\pi \times m \times \Delta v} \geq 6.06 \times 10^{-9} \text{m}$$

23. $K_1 = \frac{[AB_2]}{[AB][B_2]^{1/2}}, K_2 = \frac{[AB_2][B_2]}{[AB_2]^2}$

$$K_1^2 \cdot K_2 = \frac{[AB_2]^2}{[AB]^2 [B_2]} \cdot \frac{[AB]^2 [B_2]}{[AB_2]^2}$$

$$K_1^2 \cdot K_2 = 1$$

24. $K = \frac{\ln 2}{14.5}$

$$\frac{\ln 2}{14.5} = \frac{1}{24} \ln \left(\frac{a}{a-x} \right)$$

25. $\frac{1.26 \times 10^{-4}}{(1.26 \times 10^{-4}) + (3.8 \times 10^{-5})} = 77\% \text{ cyclohexene}$
 $\therefore \text{methyl cyclopentene} = (100 - 77) = 23\%$

27. $\frac{dE^\circ}{dT} = \frac{\Delta S^\circ}{nF}$
 $\Rightarrow \Delta S^\circ = \left(\frac{dE^\circ}{dT} \right) \times nF$
 $= -0.125 \times 2 \times 96500$
 $= -24.125 \times \text{KJ K}^{-1}$

28. $E_{\text{Cell}} = \frac{-0.0591}{2} \log \frac{1}{10} = \frac{0.0591}{2} V = 0.0295$

30. Moles escaped = original – final moles
 $= \frac{4 \times 10}{0.082 \times 290} - \frac{2.5 \times 10}{0.082 \times 290} = 0.630 \text{ mol}$

32. $\alpha = \sqrt{\frac{K_b}{C}}$
 $\Rightarrow \frac{\alpha_2}{\alpha_1} = \sqrt{\frac{C_1}{C_2}} = 2$
 $\therefore V_2 = 4 \times V_1 = 4 \times 50 = 200 \text{ ml}$

33. $\pi = \frac{CRT}{M}$
 $\frac{RT}{M} = 4.65 \times 10^{-3}$
 $8.314 \times \frac{293}{M} = 4.65 \times 10^{-3}$
 $M = 5.15 \times 10^5$

38. As only coloured salts gives coloured bead hence B, C D can not be right answers.

41. $1/8 \times 8 = 1 \text{ Au atom / unit cell}$
 $1/2 \times 6 = 3 \text{ Cu atoms/unit cell}$
 Empirical formula is AuCu_3

42. Meq. of $\text{Na}_2\text{CO}_3 \cdot x\text{H}_2\text{O}$ in 20 mL = $19.8 \times \frac{1}{10}$
 $\therefore \text{Meq. of } \text{Na}_2\text{CO}_3 \cdot x\text{H}_2\text{O} \text{ in } 100 \text{ ml} = 19.8 \times \frac{1 \times 5}{10}$
 $\therefore \frac{W}{E} \times 100 = 19.8 \times \frac{1}{2}$

$$\text{or } \frac{0.7}{\frac{M}{2}} \times 1000 = 19.8$$

$$\therefore M = 141.41$$

$$\therefore 23 \times 2 + 12 + 3 \times 16 + 18x = 141.41$$

$$\therefore x = 2$$

43. For triatomic gas

$$\alpha = 1.33, C_p = 4R$$

$$T_2 = T_1 \left(\frac{V_1}{V_2} \right)^{\gamma-1}$$

$$= 300 \times \left(\frac{5}{40} \right)^{1.33-1} = 300 \times \left(\frac{1}{8} \right)^{1/3} = 300 \times 1/2 = 150$$

$$\Delta H = n C_p (T_2 - T_1) = 3 \times 4 \times 8.314 \times (150 - 300) = -14.965 \text{ KJ}$$

44. $W = 1 \times \Delta V$

$$V_2 = \frac{R \times 373}{1} = 373 \times 0.082 = 30.58 \text{ Litre}$$

$$V_1 = 18 \text{ ml}$$

$$W = 1 \times 10^5 \times 30.58 \times 10^{-3}$$

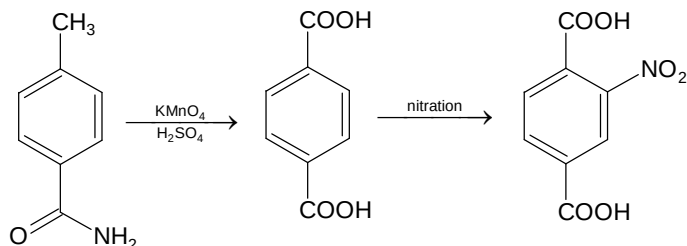
$$= 30.58 \times 10^2 \text{ J}$$

$$= 3.058 \text{ KJ}$$

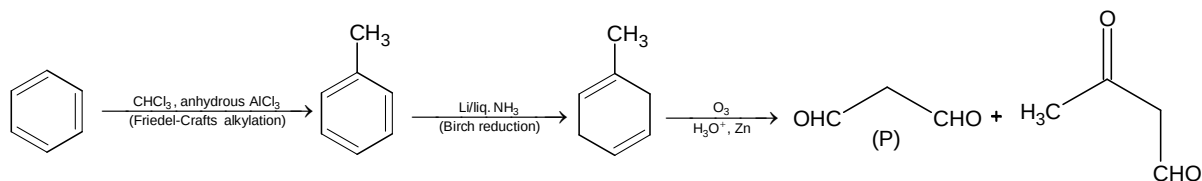
Hence (C)

$$\Delta V = V_2 - V_1 \approx V_2$$

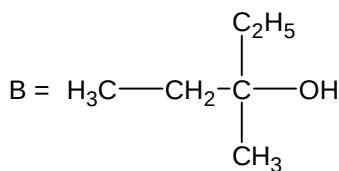
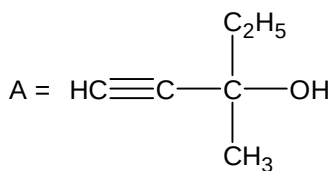
46.



47.



51.



60.

(A) Work done by the gas is ($P_1 V_1$)

(B) The maximum temperature attained by the gas during cycle $\left(\frac{25P_1 V_1}{8R} \right)$

(C) The heat absorbed by path PQ is ($3P_1 V_1$)

(D) The net heat rejected by gas in path RP is $\frac{5}{2} P_1 V_1$